

Master Project Checklist

Weakly Supervised Ideology Prediction in Social Media Text

Jacob

Billy

Shared

How to Read This Document

Every task has an **owner**, **requirements**, **acceptance criteria**, and a **definition of done**. Once every task's definition of done is met and all eight checkpoints (A–H) are signed off, the project is finished.

Ownership principle: Each person should be able to work independently most of the time. “Shared” means you need to coordinate, but one person should still be the driver. Dependencies are noted explicitly.

Task count:

- **Jacob** : 13 tasks (Reddit ingestion, core weak supervision, hierarchical model, calibration/uncertainty)
- **Billy** : 14 tasks (text cleaning, two weak sources, baselines, core transformer models, evidence/psychology heads, MITweet, eval infrastructure)
- **Shared** : 9 tasks (setup, annotation, running experiments, deliverables)

Project-Level Definition of Done

The project is done when:

1. The full data pipeline exists for Reddit weak supervision, gold annotation, and MITweet transfer.
2. The model ladder B1 through B6 is implemented and evaluated.
3. The final model includes target, stance, frame, psychological cues, evidence spans, soft-label fine-tuning, calibration, abstention, and consistency regularization.
4. All four evaluation regimes are run: in-domain, leave-one-topic-out, leave-one-community-out, and external transfer to MITweet.
5. All key metrics, ablations, robustness analysis, counterfactual consistency tests, and structured error analysis are complete.
6. Checkpoints A through H are satisfied.

1 Foundations (Tasks 1–4)

Shared — Task 1: Freeze the task definition and label space

Requirement. Lock the exact prediction targets so nobody builds mismatched datasets or heads. Level A: relevance, target, stance, frame, moralization, identity signaling, evidence spans. Level B: economic direction, social direction, intensity, ambiguity.

Acceptance criteria.

- One written spec with every label, class set, and allowed value.

- MITweet label mapping documented.
- Both Jacob and Billy sign off.

Definition of done. A single versioned task-spec document exists, both agree to it, and all downstream code references that spec only.

Shared — Task 2: Set up repo, environment, and database

Requirement. Python pipeline, PyTorch + Hugging Face, scikit-learn, PostgreSQL with schema: `raw_posts`, `user_priors`, `gold_annotations`, `gold_aggregates`, `model_outputs`, `eval_results`.

Acceptance criteria.

- Reproducible env setup from a fresh clone.
- PostgreSQL running, all tables created.
- Smoke test: insert and query one row from every table.

Definition of done. Either person can clone, set up, and run a smoke test without help.

Suggested split. Jacob: database schema and migrations. Billy: repo structure, environment files, training/eval scaffold.

Jacob — Task 3: Build Reddit ingestion pipeline

Requirement. Collect ~50K+ posts from politically relevant subreddits with text, subreddit, user ID, flair, timestamp, and thread context.

Acceptance criteria.

- Collection script runs end to end.
- All required fields present.
- Subreddit list documented.
- Records stored in `raw_posts`.

Definition of done. Nontrivial corpus queryable by subreddit, user, and timestamp without missing fields.

Billy — Task 4: Clean, normalize, and deduplicate text

Requirement. Checkpoint A requires no duplicates, no broken text.

Dependency. Operates on the raw corpus from Jacob's Task 3.

Acceptance criteria.

- Duplicate detection implemented.
- Broken/empty text filtered.
- Basic normalization documented.
- Data-quality report with before/after counts.

Definition of done. Checkpoint A data hygiene satisfied; saved QA summary proves it.

2 Weak Supervision Pipeline (Tasks 5–12)

Jacob — Task 5: Weak source 1 — flair / self-declaration priors

Requirement. Parse user flair, map to economic/social priors with confidence.

Acceptance criteria. Flair parsing rules exist. Mapping documented. Source confidence assigned.

Definition of done. Given a recognized flair, the system produces valid econ/social scores plus confidence.

Billy — Task 6: Weak source 2 — community-participation priors

Requirement. Frequency-weighted prior from user’s subreddit posting distribution using a curated community-to-axis mapping.

Acceptance criteria. Community-to-axis mapping file exists. Aggregation works. Priors stored in `user_priors`.

Dependency. Uses the raw user/post data from Jacob’s Task 3.

Definition of done. Any user with sufficient activity gets a reproducible community-based prior.

Billy — Task 7: Weak source 3 — self-identification phrase rules

Requirement. Detect “I’m conservative,” “as a socialist,” etc. and map to axis-specific priors.

Acceptance criteria. Pattern library exists. Mapped priors produced with high-confidence tags.

Dependency. Uses the raw post text from Jacob’s Task 3. Output feeds into Jacob’s aggregation (Task 11).

Definition of done. Validation sample shows rules fire correctly on explicit self-ID language.

Jacob — Task 8: Weak source 4 — target/frame lexical cues

Requirement. Seed sets for political targets and framing cues, context-gated (not naive keyword).

Acceptance criteria. Target and frame seed lexicons exist. Context gating implemented. Rules do not fire on keyword alone.

Definition of done. Examples show generic keyword matches blocked unless context supports intended reading.

Jacob — Task 9: Contextualized rule firing

Requirement. Every weak source emits econ/social scores and confidence; firing is context-aware.

Acceptance criteria. Context-aware activation implemented. False-positive tracking. Outputs: z_{econ} , z_{soc} , $c_s(x)$.

Definition of done. Validation report shows better precision than naive keyword firing.

Jacob — Task 10: Source-reliability estimation

Requirement. Reliability depends on source-type prior, contextual match, inter-source agreement, historical accuracy.

Acceptance criteria. Reliability function implemented. Calibration subset logic exists. Outputs inspectable per example.

Definition of done. Given a post, the system produces reliability weights $r_s(x)$ with audit trail.

Jacob — Task 11: Soft-label aggregation and quality scoring

Requirement. Denoised soft labels for econ/social direction plus quality score $q(x)$.

Acceptance criteria. Aggregated soft labels produced. Entropy computed. $q(x)$ computed and clipped. Low-quality examples filterable.

Definition of done. Every weakly labeled Reddit example has valid soft targets and a saved quality score.

Jacob — Task 12: Target-conditioned weak projection

Requirement. Weak labels flow through target/stance/frame hypotheses before projecting to ideology.

Acceptance criteria. Weak target, stance, frame hypotheses exist. Rule-based projection $g(\dots)$ implemented.

Definition of done. Target→stance→frame→ideology weak pipeline runs with inspectable intermediate outputs.

3 Gold Annotation (Tasks 13–14) — Shared

Shared — Task 13: Build annotation tool, guide, and sampling plan

Requirement. 1,000–1,500 posts, stratified by subreddit/topic/weak-prior quartile, 3 annotators, 12 questions including evidence spans.

Acceptance criteria. Annotation instructions exist. Sampling strategy implemented. Evidence span capture supported.

Definition of done. An annotator can receive the guide and tool and produce a valid full annotation vector.

Suggested split. Jacob: sampling strategy and stratification code. Billy: annotation interface/spreadsheet design and evidence-span capture mechanism.

Shared — Task 14: Collect and validate the gold set

Requirement. 3 annotations per post. Majority labels, soft distributions, disagreement entropy computed.

Acceptance criteria. Full annotation collected. Distributions computed. Data stored in `gold_annotations` and `gold_aggregates`.

Definition of done. Full target-size gold set exists, distributions are correct, Checkpoint A annotation side satisfied.

4 Baselines & Core Transformer Models (Tasks 15–20) — Billy

Billy — Task 15: B1 lexical baseline (TF-IDF + LR)

Requirement. Sanity baseline. Checkpoint B: beats random, class imbalance understood.

Acceptance criteria. TF-IDF + LR trains on gold labels. Metrics reported. Random baseline comparison exists.

Definition of done. Checkpoint B satisfied: B1 beats random.

Billy — Task 16: B2 flat transformer baseline

Requirement. RoBERTa-base on gold hard labels, flat heads for relevance/econ/social.

Acceptance criteria. Loss curves normal. In-domain beats B1. No label leakage.

Definition of done. Checkpoint C satisfied: transformer beats lexical baseline.

Billy — Task 17: B3 weak-pretrained transformer (hard fine-tune)

Requirement. Take Jacob's weak-label dataset (from Tasks 5–12), pretrain RoBERTa, then fine-tune on gold hard labels.

Dependency. Jacob must deliver the weak-label dataset first (Task 11 output).

Acceptance criteria. Weak pretraining runs on Reddit corpus. Fine-tuning on gold set works. Results compared to B2.

Definition of done. B3 checkpoint exists. Weak pretraining shows improvement over B2 on at least one shift setting (Checkpoint D).

Billy — Task 18: B4 weak-pretrained transformer (soft fine-tune)

Requirement. Same as B3 but fine-tune with soft annotator-distribution labels instead of hard majority.

Dependency. Requires gold soft labels from Task 14.

Acceptance criteria. Soft-label loss (KL divergence) implemented. B4 evaluated against B3.

Definition of done. B4 checkpoint exists. Soft labels show measurable benefit on ambiguity-heavy examples.

Billy — Task 19: B5 flat multi-task model

Requirement. Extend B4 with ambiguity and framing heads (still flat architecture — no hierarchy). This is the critical comparator for B6.

Acceptance criteria. Flat multi-head architecture includes ambiguity and framing. Calibration pipeline exists. B5 results reported against B4.

Definition of done. Fully working flat multi-task model, strong enough to make B5 vs. B6 a meaningful comparison.

Billy — Task 20: MITweet ingestion, preprocessing, and label mapping

Requirement. MITweet is the primary external benchmark for Regime D.

Acceptance criteria. MITweet loader works. Clean train/val/test access. Relevance and ideology facet mapping documented.

Definition of done. MITweet usable as external benchmark. Mapping assumptions written down.

5 Hierarchical Model & Uncertainty (Tasks 21–26) — Jacob**Jacob — Task 21: B6 hierarchical model architecture**

Requirement. Level A heads (relevance, target, stance, frame) feed intermediate representations into a projection layer; Level B heads (econ, social, intensity, ambiguity) consume the projection. Latent ideological vector $\mathbf{z}(x)$ with orthogonality regularization.

Acceptance criteria. Shared encoder implemented. Target, stance, frame heads produce intermediate representations. Ideology projection uses intermediates, not raw encoder alone. Orthogonality regularization implemented.

Definition of done. Full hierarchical forward pass produces all required outputs for one batch without stubs.

Jacob — Task 22: Soft-label fine-tuning and disagreement-aware training

Requirement. Gold-stage training on annotator distributions with KL loss. Ambiguity head active.

Acceptance criteria. KL soft-label objective used for ideology heads. Ambiguity head exists. Ambiguity-heavy subset evaluation exists.

Definition of done. Checkpoint F satisfied: soft-label training improves ambiguity-heavy behavior.

Jacob — Task 23: Consistency regularization

Requirement. User-level temporal consistency loss during weak pretraining.

Acceptance criteria. Pairwise user-post consistency loss runs. Can be toggled for ablation.

Definition of done. Training logs prove it is active; ablation toggle works cleanly.

Jacob — Task 24: Confidence-based curriculum

Requirement. Time-varying quality weights during weak pretraining: clean examples first, noisy examples later.

Acceptance criteria. Curriculum schedule implemented. Can be toggled for ablation.

Definition of done. Training logs show the curriculum is active (early epochs weight high- q examples more).

Jacob — Task 25: Calibration and abstention

Requirement. Post-hoc temperature scaling or isotonic regression. Abstention threshold for uncertain cases.

Acceptance criteria. Calibration fit runs. ECE and Brier computed. Risk-coverage analysis exists.

Definition of done. Checkpoint H satisfied: calibrated confidence better than uncalibrated; abstention reduces error.

Jacob — Task 26: Continuous signal formulation and vector field analysis

Requirement. Latent vector projections onto interpretable axes. Community-level velocity fields with divergence/curl.

Acceptance criteria. Continuous scores exported. Community-level means over time computed. At least one visualization.

Definition of done. Analysis notebook showing ideological drift/polarization geometry on model outputs.

6 Evidence & Psychology Model Extensions (Tasks 27–28) — Billy**Billy — Task 27: Evidence-span extraction head**

Requirement. Token-level classifier predicting whether each token belongs to an evidence span. Produces evidence-weighted representation. Serves interpretability, denoising, and error analysis.

Dependency. Integrates into Jacob’s B6 architecture (Task 21). Billy builds the head; Jacob integrates it into the hierarchical forward pass.

Acceptance criteria. Token-level evidence head runs. BCE loss implemented. Token P/R/F1 computed. Evidence-weighted representation can be toggled on/off.

Definition of done. Model predicts evidence spans, you can inspect them on examples, and evidence metrics are part of eval. Checkpoint G (evidence plausibility) is addressed.

Billy — Task 28: Psychological cue heads (moralization + identity signaling)

Requirement. Two sub-heads predicting moralization (low/med/high) and identity signaling (low/med/high). Their penultimate outputs feed into the psychology representation $\mathbf{p}(x)$.

Dependency. Integrates into Jacob’s B6 architecture. Billy builds the heads; Jacob wires $\mathbf{p}(x)$ into the projection layer.

Acceptance criteria. Both sub-heads produce distributions. Labels exist in gold set. Heads can be toggled for ablation.

Definition of done. Psychology heads train, produce outputs, and their ablation (on/off) is runnable.

7 Evaluation Infrastructure (Tasks 29–31)**Billy — Task 29: Define and implement all evaluation splits**

Requirement. Four regimes: in-domain, leave-one-topic-out, leave-one-community-out, external transfer.

Acceptance criteria. In-domain split fixed. Topic partitions fixed. Community partitions fixed. External transfer pipeline fixed. Leakage checks documented.

Definition of done. A script reproduces every split exactly; no post in incompatible partitions.

Billy — Task 30: Implement all metrics

Requirement. Standard (macro-F1, micro-F1, balanced acc), ordinal (MAE, QWK), disagreement-aware (KL, JSD), calibration (ECE, Brier), selective prediction (risk-coverage), signal-level (Pearson, Spearman),

decomposition (target/stance/frame F1, ideology conditioned on correct target/stance), evidence (token P/R/F1).

Acceptance criteria. One eval script produces all required metrics for a given model and split.

Definition of done. Script is tested on B1 outputs and produces a complete metric bundle.

Billy — Task 31: Build counterfactual edit set

Requirement. Four edit types: target swap, stance reversal, frame change, cue removal. ~200 edited posts.

Acceptance criteria. Counterfactual set exists with all four types. Direction of expected change documented per edit.

Definition of done. Counterfactual set is ready for evaluation on any model.

8 Running Experiments (Tasks 32–35) — Shared

Shared — Task 32: Run the full model ladder and produce comparison tables

Requirement. B1 through B6 across all four regimes. B5 vs. B6 is the critical comparison.

Suggested split. Billy runs B1–B5 evaluations. Jacob runs B6 evaluations. Both review the combined table.

Definition of done. Clean master results table; you can explain what each step in the ladder added.

Shared — Task 33: Run the ablation suite (A1–A13)

Requirement. 13 ablations. The most important is A7 (flat vs. hierarchical).

Suggested split.

- Billy runs: A1 (weak vs. no weak), A2 (denoised vs. raw), A3 (hard vs. soft), A9 (evidence on/off), A10 (psych on/off). *These ablations involve Billy's models.*
- Jacob runs: A4 (ambiguity on/off), A5 (target/frame sensitivity), A6 (calibration on/off), A7 (flat vs. hierarchical), A8 (oracle target/stance), A11 (consistency on/off), A12 (curriculum on/off), A13 (counterfactual flat vs. hierarchical). *These ablations involve Jacob's models.*

Definition of done. Every major claim has an ablation backing it up.

Shared — Task 34: Run robustness analysis and counterfactual evaluation

Requirement. Bootstrap CIs, noise perturbation (5/10/20%), distribution reweighting, counterfactual consistency.

Suggested split. Jacob: bootstrap CIs and noise perturbation. Billy: distribution reweighting and counterfactual consistency scoring (using the edit set from Task 31).

Definition of done. Robustness appendix shows gains are not split luck. Counterfactual table shows hierarchical model is more compositional.

Shared — Task 35: Structured error analysis

Requirement. Stage-decomposed (target/stance/frame/projection failure) plus content-based taxonomy (sarcasm, vague, identity, mixed, etc.). 100+ examples.

Suggested split. Each person examines 50+ errors independently, then merge and reconcile categories.

Definition of done. Error analysis explains not only how often the model fails but where and why.

9 Final Deliverables (Task 36) — Shared

Shared — Task 36: Produce all final deliverables and validate checkpoints

Requirement. Reproducible artifacts and publishable outputs.

Deliverables.

- Clean repo with README
- Reproducible data pipeline docs
- Annotation guide
- Main results tables (model ladder $\times$ 4 regimes)
- Ablation table
- Decomposition analysis table
- Counterfactual consistency table
- Robustness appendix (bootstrap CIs, noise sensitivity)
- Error analysis with figures
- Final paper draft

Checkpoint validation.

A: Clean data, plausible priors, annotation works

B: B1 beats random

C: B2 beats B1, no leakage

D: Weak pretraining helps somewhere

E: Hierarchical matches or beats flat under shift

F: Soft labels help ambiguity-heavy cases

G: Evidence spans are plausible

H: Calibration improves confidence quality

Suggested writing split.

- Jacob writes: Abstract, Introduction, Weak Supervision (Sec. 5), Hierarchical Model (Sec. 6), Calibration/Uncertainty.
- Billy writes: Related Work, Data (Sec. 4), Baselines & Core Models, Experimental Setup (Sec. 7), Evidence/Psychology analysis.
- Shared: Results, Error Analysis, Conclusion, proofreading.

Definition of done. A third person could reproduce core experiments, understand the setup, and see evidence for every claim.

Ownership Summary

#	Task	Owner	Checkpoint
1	Freeze task definition	Shared	—
2	Repo, env, database	Shared	—
3	Reddit ingestion	Jacob	A
4	Text cleaning	Billy	A
5	Weak source 1: flair	Jacob	A
6	Weak source 2: community	Billy	A
7	Weak source 3: self-ID	Billy	A
8	Weak source 4: target/frame	Jacob	A
9	Contextualized rule firing	Jacob	A
10	Source reliability estimation	Jacob	—
11	Soft-label aggregation + quality	Jacob	—
12	Target-conditioned weak projection	Jacob	—
13	Annotation tool + guide	Shared	A
14	Collect gold annotations	Shared	A
15	B1: TF-IDF + LR	Billy	B
16	B2: Flat transformer	Billy	C
17	B3: Weak + hard fine-tune	Billy	D
18	B4: Weak + soft fine-tune	Billy	D
19	B5: Flat multi-task	Billy	—
20	MITweet ingestion + mapping	Billy	—
21	B6: Hierarchical model	Jacob	E
22	Soft-label / disagreement training	Jacob	F
23	Consistency regularization	Jacob	—
24	Confidence-based curriculum	Jacob	—
25	Calibration + abstention	Jacob	H
26	Signal formulation + vector fields	Jacob	—
27	Evidence-span head	Billy	G
28	Psychology heads	Billy	—
29	Evaluation splits	Billy	—
30	All metrics script	Billy	—
31	Counterfactual edit set	Billy	—
32	Run model ladder	Shared	—
33	Run ablation suite	Shared	—
34	Robustness + counterfactual eval	Shared	—
35	Error analysis	Shared	—
36	Final deliverables + checkpoints	Shared	A–H

Table 1: Complete task ownership summary. Jacob: 13 tasks. Billy: 14 tasks. Shared: 9 tasks.

	Jacob's Track	Billy's Track
Data	Reddit ingestion, weak sources 1 & 4, contextualized firing, reliability, aggregation, weak projection (7 tasks)	Text cleaning, weak sources 2 & 3, MITweet pipeline (4 tasks)
Models built	B6 hierarchical + calibration + curriculum + consistency (6 tasks)	B1, B2, B3, B4, B5 + evidence head + psychology heads (8 tasks)
Eval infra	—	Splits, metrics, counterfactual set (3 tasks)
Analysis	Vector fields, signal formulation (1 task)	—
Integration	Tasks 32–36 shared	

Table 2: Track summary. Jacob: 13 tasks. Billy: 14 tasks. Shared: 9 tasks. Billy builds and evaluates 7 models (B1–B5 + evidence + psychology), builds two weak-supervision sources, and owns all evaluation infrastructure. Jacob builds the Reddit pipeline, core denoising logic, and the final hierarchical model with uncertainty.

Critical Dependencies

The two tracks are mostly independent, but these handoffs matter:

1. **Jacob → Billy (raw data).** Billy needs the cleaned Reddit corpus (Tasks 3–4 output: Jacob ingests, Billy cleans) to build community-participation priors (Task 6) and self-ID phrase rules (Task 7). Jacob should deliver raw data early in Week 1.
2. **Billy → Jacob (weak sources 2 & 3).** Jacob's aggregation and quality scoring (Task 11) consumes outputs from *all four* weak sources. Billy must deliver community priors (Task 6) and self-ID priors (Task 7) before Jacob can finalize the denoised weak-label dataset.
3. **Jacob → Billy (weak-label dataset).** Billy needs the final weak-label dataset (Task 11 output) to train B3 and B4 (Tasks 17–18). Jacob should deliver this as soon as aggregation is done.
4. **Billy → Jacob (B5 checkpoint).** Jacob needs a working B5 (Task 19) to run the B5-vs-B6 comparison. Billy should deliver B5 before Jacob's B6 is finalized.
5. **Billy → Jacob (evidence + psychology heads).** Billy builds the evidence head (Task 27) and psychology heads (Task 28) as modules. Jacob integrates them into B6's forward pass.
6. **Both → evaluation.** Tasks 32–36 cannot start until both tracks have trained models. Plan a hard sync point before the evaluation phase.